

# Suyash Soni

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## SUMMARY

AI/Backend Engineering student who designs and ships **end-to-end backend systems for AI applications** — REST APIs, LLM integration, and real-time data pipelines using FastAPI, PostgreSQL, and Groq LLM APIs. Delivered production-style projects independently from architecture through deployment, with measurable model and system performance. Currently extending this into infra (Docker, CI/CD) and applied LLM engineering (RAG, agentic workflows).

## TECHNICAL SKILLS

**Backend & APIs:** FastAPI, REST APIs, PostgreSQL, SQLite, JWT Authentication, Node.js

**LLM / GenAI Integration:** Groq LLaMA 3.1, LLM API Integration, Prompt Engineering, CLIP Visual Search, Huggingface

**Computer Vision:** YOLO11, YOLO8, YOLO-World, DeepSORT, OpenCV, PyTorch, EfficientNet, DeepLabV3+

**Languages:** Python, C, C++, Java, JavaScript (basic), HTML (basic)

**Data & Analytics:** NumPy, Pandas, Matplotlib, SQL, MongoDB, Tableau, Jupyter Notebook

**Developer Tools:** Git, VS Code, Kaggle, Google Colab, Vercel, Figma (basic)

## EXPERIENCE

**Full Stack Web Development Intern** | [Github link\(fork\)](#)

Jun 2025 – Aug 2025

*Techbrill Solutions*

*Remote*

- **Delivered** 50+ reusable UI components adopted across client projects, cutting duplicate development effort for the team
- **Maintained** a consistent 50+ commit delivery pace over a 1-month sprint, demonstrating reliable, self-directed execution in a remote setting
- **Optimized** website performance, directly improving client-facing user experience metrics

## PROJECTS

**Visual Memory Lane** | [Github link](#)

May 2026

*FastAPI, PostgreSQL, Groq LLaMA 3.1, YOLO11, DeepSORT, React Native*

*Personal Project*

- **Architected** a **FastAPI** backend with **PostgreSQL**, JWT authentication, and CLIP-based visual search, supporting a **camera-agnostic** AI assistant across up to 4 simultaneous live feeds
- **Built** an LLM reasoning layer with Groq LLaMA 3.1 to convert raw detection events into scene-aware suggestions, interval summaries, and natural-language query answers — reducing manual review of camera history to a single chat query
- **Implemented** real-time object detection and per-ID tracking (**YOLO11 + DeepSORT**) with class persistence across frames, feeding structured events directly into the backend pipeline
- **Shipped** a React Native mobile client giving end-to-end cross-platform access to object history, summaries, and live monitoring — full-stack ownership from model to mobile UI

**LedgerApp** | **Native Android Billing System** | [Github link](#)

*Kotlin, Room Database, MVVM, Jetpack (ViewModel, Navigation)*

*Personal Project*

- **Built** an offline-first transaction management app for Indian retail shopkeepers, prioritizing accessibility for low digital-literacy users
- **Implemented** MVVM architecture with Room (KSP) for local persistence, eliminating dependency on network connectivity for core billing operations
- **Resolved** a Gradle/AGP 9.2.1 + Kotlin toolchain incompatibility by migrating the build system from `kapt` to KSP, unblocking the entire build pipeline

**Road Network Segmentation from Satellite Imagery** | [Github link](#)

March 2026

*Python, PyTorch, DeepLabV3+, EfficientNet-B4, OpenCV*

*Academic Project*

- **Built** a road segmentation system using DeepLabV3+ with an EfficientNet-B4 backbone, trained on 5,634 annotated Sentinel-2 image-mask pairs with mixed-precision GPU training
- **Achieved** F1 **0.5136** and IoU **0.3469** on held-out test data (exceeding target thresholds of  $F1 \geq 0.50$  and  $IoU \geq 0.38$ ), with **62.6% recall** via a composite Dice/Focal/BCE loss and ASPP multi-scale convolutions

## EDUCATION

**Vellore Institute of Technology (VIT) Bhopal**

Bhopal, MP

*B.Tech in Computer Science Engineering (AI & ML)*

*Expected July 2027*

- CGPA: 8.27

## ACHIEVEMENTS AND CERTIFICATIONS

**Internship Completion Letter** - TechBrill Solutions | 3 months

[View Certificate](#)

**Applied Machine Learning in Python** - University of Michigan | 31 Hours

[View Certificate](#)

**Cloud Computing (Elite Certificate)** - NPTEL, IIT Kharagpur | 64% Score, Top Percentile

[View Certificate](#)

**Geodata Processing using Python** - IIRS, ISRO | 7.5 Hours, 100% Completion

[View Certificate](#)